

Muhammad Fayaz Khan

LLM Engineer | RAG Systems • Agentic AI • FastAPI • LangChain

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PROFESSIONAL SUMMARY

LLM Engineer specializing in Enterprise-grade RAG Systems and Agentic AI workflows. Experienced in building production-ready AI applications using FastAPI, LangChain, and Vector Databases with a focus on retrieval accuracy, latency optimization, and LLM cost reduction. Proven ability to design end-to-end AI pipelines from chunking strategy to RAGAS evaluation with measurable results.

TECHNICAL SKILLS

Generative AI & RAG	RAG Pipeline, Hybrid Search (Dense + BM25), Neural Reranking, Self-Querying, RAGAS Evaluation
Prompt Engineering	Zero-shot, Few-shot, Chain-of-Thought, ReAct Pattern, Context Engineering
Agentic AI	LangGraph, CrewAI, LangChain, Tool Calling, Multi-Agent Workflows, MCP
Backend & APIs	FastAPI, Python (Async/Await, Decorators, Generators), Pydantic v2, Instructor Library
Vector Databases	ChromaDB, Qdrant, Pinecone
Core AI / ML	Machine Learning, Deep Learning (PyTorch), CNNs, Scikit-Learn
DevOps & Tools	Docker, Docker Compose, Git/GitHub, Streamlit, AWS
LLM Providers	OpenAI, Anthropic Claude, Google Gemini, Groq

PROJECTS PORTFOLIO

BizBrain — Business PDF Agent *[FastAPI · ChromaDB · RAGAS · RAG · LangChain · Pydantic · Chainlit · Docker]*

Enterprise FAQ agent that ingests business PDFs and answers domain-specific queries using Hybrid Search and Cross-Encoder Reranking. Designed for real client deployment with persistent chat history.

Enterprise RAG Platform *[FastAPI · ChromaDB · RAGAS · LangChain]*

Built a production-grade RAG system using Recursive Semantic Chunking and Cross-Encoder Reranking for querying private PDF data. Achieved high faithfulness and answer relevancy scores via RAGAS evaluation framework.

Multi-Functional AI Agent *[LangChain · Pydantic v2 · Tool Calling]*

Developed an autonomous agent with structured Tool Calling and JSON outputs (Pydantic) to handle complex, multi-step business workflows autonomously.

Rossmann Sales Predictor *[Python · Scikit-Learn · Feature Engineering]*

Predictive model for Rossmann retail chain sales forecasting. Applied advanced feature engineering and ensemble methods to achieve competitive accuracy on time-series retail data.

Churn Predictor *[Python · Scikit-Learn · Pandas]*

ML classification model to predict customer churn. Includes data preprocessing, feature selection, and model evaluation with precision/recall analysis.

Diabetes Risk Analyzer *[Python · Scikit-Learn · Streamlit]*

Deployed a classification model as an interactive web app for diabetes risk prediction based on patient health parameters.

House Price Prediction *[Python · Regression · Scikit-Learn]*

End-to-end regression pipeline for predicting house prices with data cleaning, EDA, and model comparison.

Cat vs Dog CNN Classifier *[PyTorch · CNN · Streamlit]*

Deep Learning image classification model using Convolutional Neural Networks, deployed as an interactive Streamlit app.

KEY COMPETENCIES

- Optimizing LLM API costs and latency for production-ready applications
- Implementing LLM Caching (GPTCache) - reducing API costs by up to 40%
- Building stateful, cyclical multi-agent workflows using LangGraph
- Designing RAG 2.0 pipelines with Hybrid Search and Neural Reranking for enterprise clients
- Integrating multiple LLM providers: OpenAI, Anthropic, Gemini, and Groq
- Red teaming and evaluating LLM outputs with RAGAS and LLM-as-Judge frameworks

EDUCATION

12th Grade (Intermediate Computer Science) — Completed

Self-Taught LLM Engineering Roadmap (2026)